

Coding Round Practice Set

20 Scenario-Based Questions Frequently Asked in
Service-Based Company Placement Tests

(TCS NQT · Infosys · Wipro Elite NLTH · Cognizant GenC · Accenture · Capgemini style)

How to use this set:

Each question below is written in the same story/scenario style used by service-based companies (TCS, Infosys, Wipro, Cognizant, Accenture, etc.) in their online coding assessments. Every question includes the problem statement, input/output format, constraints, and at least one sample input-output pair, exactly as they would appear in a real test.

Topics covered: Strings, Arrays, Hashing, Math, Recursion, Sorting, Searching (Binary Search), Stacks, 2D Arrays / Matrices, Pattern Printing, and basic Simulation — the topics that appear most often across these companies' coding rounds.

Tip: Try to solve each problem within 15–20 minutes using your language of choice (C, C++, Java, or Python), then check the sample output before moving to the next one.

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Questions

Q1

Palindrome Product Code Checker

Topic: Strings | Difficulty: Easy

BrightPack Logistics ships products across the country. The warehouse team recently found that a scanner glitch sometimes enters a product code back-to-front. Since a palindrome code reads identically forwards and backwards, the QA team wants a quick check built into the scanning software: given a product code, determine whether it is a palindrome (ignore case).

Input Format:

A single line containing a string S of length 1 to 100 (letters and digits only).

Output Format:

Print "PALINDROME" if S reads the same forwards and backwards (case-insensitive); otherwise print "NOT A PALINDROME".

Constraints:

$1 \leq |S| \leq 100$

Sample Input

MADAM

Sample Output

PALINDROME

Sample Input 2

Hello123

Sample Output 2

NOT A PALINDROME

Q2**Anagram Checker for a Word-Puzzle App**

Topic: Strings / Hashing | Difficulty: Easy

WordWhiz, a mobile word-puzzle game, lets two players build words from the same set of scrambled letter tiles. To award a special “Twin Words” badge, the app must check whether the two words submitted are anagrams of each other — containing exactly the same letters with the same frequency, ignoring case and spaces.

Input Format:

Two lines, each containing one word/phrase.

Output Format:

Print "ANAGRAM" if the two inputs are anagrams of each other, else print "NOT ANAGRAM".

Constraints:

$1 \leq \text{length of each string} \leq 1000$

Sample Input

```
listen  
silent
```

Sample Output

```
ANAGRAM
```

Sample Input 2

```
hello  
world
```

Sample Output 2

```
NOT ANAGRAM
```

Q3**Second-Highest Sales – SuperMart Regional Report**

Topic: Arrays | Difficulty: Easy

SuperMart's regional manager receives a monthly report with the sales figures of every branch store. Before announcing the “Runner-Up Store” bonus, she needs the second-highest DISTINCT sales value from the list. Help her by writing a program that finds it. You may assume the list has at least two distinct values.

Input Format:

First line: integer n (number of stores). Second line: n space-separated integers (sales figures).

Output Format:

Print the second-highest distinct sales value.

Constraints:

$2 \leq n \leq 10^5$, $0 \leq \text{sales}[i] \leq 10^9$

Sample Input

```
5  
45000 67000 67000 52000 39000
```

Sample Output

```
52000
```

Q4**Most Reported Issue – IT Helpdesk Analytics**

Topic: Hashing | Difficulty: Easy-Medium

The IT helpdesk at Nova Solutions logs every support ticket with a numeric issue-category code. At the end of the week, the support-ops lead wants to know which category was reported the most so the team can prioritize a fix first. If two categories are tied for the highest count, report the smaller category code.

Input Format:

First line: integer n . Second line: n space-separated integers (category codes).

Output Format:

Print the most frequently occurring category code.

Constraints:

$1 \leq n \leq 10^5$

Sample Input

```
7
101 102 101 103 102 101 104
```

Sample Output

```
101
```

Q5**Gate Access Code Validator**

Topic: Math | Difficulty: Easy

A corporate campus has installed a smart security gate that issues every visitor a numeric access code. As an extra layer of protection, the system is designed to only accept codes that are prime numbers. Write a program that reads a number and reports whether it is a valid access code.

Input Format:

A single integer N .

Output Format:

Print "VALID" if N is prime, otherwise print "INVALID".

Constraints:

$1 \leq N \leq 10^6$

Sample Input

```
29
```

Sample Output

```
VALID
```

Sample Input 2

```
15
```

Sample Output 2

```
INVALID
```

Q6**Task Ordering Possibilities – Workshop Scheduler**

Topic: Math / Recursion | Difficulty: Easy

A manufacturing workshop has N independent tasks to finish before the shift ends. The shift supervisor wants to know, purely for a randomized-testing routine, in how many different orders the N tasks could theoretically be arranged. This count is simply $N!$ (N factorial).

Input Format:

A single integer N .

Output Format:

Print the value of $N!$ (factorial of N).

Constraints:

$0 \leq N \leq 20$

Sample Input

5

Sample Output

120

Q7**Bacterial Growth Simulation – BioLab Research**

Topic: Recursion / DP | Difficulty: Easy

Researchers at a biology lab are modelling a hypothetical bacterial colony whose population count at each hour follows the Fibonacci sequence, starting from 0 and 1. To plot a growth chart, they need the population values for the first N hours printed in order.

Input Format:

A single integer N (number of terms).

Output Format:

Print the first N Fibonacci numbers, space-separated, on one line.

Constraints:

$1 \leq N \leq 40$

Sample Input

8

Sample Output

0 1 1 2 3 5 8 13

Q8**Armstrong Serial Certification – QuantumTech Electronics**

Topic: Math | Difficulty: Easy

QuantumTech's quality-assurance team stamps a special "Certified" badge on any device whose numeric serial number is an Armstrong number — a number equal to the sum of each of its digits raised to the power of the total digit count. Write a program that checks this for a given serial number.

Input Format:

A single integer N.

Output Format:

Print "ARMSTRONG NUMBER" if N qualifies, else print "NOT AN ARMSTRONG NUMBER".

Constraints:

$1 \leq N \leq 10^8$

Sample Input

153

Sample Output

ARMSTRONG NUMBER

Sample Input 2

120

Sample Output 2

NOT AN ARMSTRONG NUMBER

Q9**Image Preview Rotation – Matrix Transpose**

Topic: 2D Arrays | Difficulty: Easy

The engineering team at PixelCraft, a photo-editing startup, is building a "quick rotate preview" feature. As the first step of their 90-degree rotation pipeline, they need to transpose the pixel-intensity matrix of an image — that is, turn its rows into columns. Given a matrix of R rows and C columns, print its transpose.

Input Format:

First line: two integers R and C. Next R lines: C space-separated integers each (the matrix).

Output Format:

Print the transposed matrix (C rows, R columns), each row on a new line.

Constraints:

$1 \leq R, C \leq 100$

Sample Input

2 3
1 2 3
4 5 6

Sample Output

1 4
2 5
3 6

Q10**Leaderboard Ranking – GameZone Arena**

Topic: Sorting | Difficulty: Easy

GameZone Arena, an online multiplayer gaming platform, displays a live leaderboard after every match round, ordered from the highest score to the lowest so that top performers are shown first. Write a program that sorts a list of player scores in descending order.

Input Format:

First line: integer n. Second line: n space-separated integers (scores).

Output Format:

Print the scores sorted in descending order, space-separated.

Constraints:

$1 \leq n \leq 10^5$

Sample Input

```
5
34 89 12 76 50
```

Sample Output

```
89 76 50 34 12
```

Q11**Employee ID Lookup – Zenith Corp HR Portal**

Topic: Searching | Difficulty: Easy-Medium

The HR portal of Zenith Corp stores all employee IDs in sorted order to enable fast lookups. When an HR executive searches for a specific employee ID, the system should use binary search to instantly report the 1-indexed position at which it exists, or state that it was not found.

Input Format:

First line: integer n. Second line: n sorted space-separated integers (employee IDs). Third line: the target ID to search for.

Output Format:

Print "Found at position X" (1-indexed) if present, otherwise print "NOT FOUND".

Constraints:

$1 \leq n \leq 10^6$, IDs are sorted in strictly increasing order

Sample Input

```
6
101 105 110 118 125 130
118
```

Sample Output

```
Found at position 4
```

Q12 Review Readability Analyzer

Topic: Strings | Difficulty: Easy

An e-commerce company's content-quality team is building a review analyzer. As a first step toward a readability score, they want to count how many vowels and how many consonants appear in a given customer review (spaces are ignored, and the letter 'y' is treated as a consonant).

Input Format:

A single line of text containing only letters and spaces.

Output Format:

Print the counts in the exact format: "Vowels: X, Consonants: Y".

Constraints:

$1 \leq \text{length of text} \leq 1000$

Sample Input

```
Great product very useful
```

Sample Output

```
Vowels: 8, Consonants: 14
```

Q13 Code Syntax Checker – Balanced Brackets

Topic: Stack | Difficulty: Medium

A startup building a lightweight browser-based code editor wants a “quick syntax check” feature. Before a user hits compile, the editor should verify that every opening bracket — (), {}, and [] — in the code snippet has a matching closing bracket in the correct order.

Input Format:

A single line containing only the characters (){}[].

Output Format:

Print "BALANCED" if the brackets are correctly matched and nested, else print "NOT BALANCED".

Constraints:

$1 \leq \text{length of string} \leq 10^5$

Sample Input

```
{[()()]}
```

Sample Output

```
BALANCED
```

Sample Input 2

```
{[()]}
```

Sample Output 2

```
NOT BALANCED
```

Q14**Reverse Mode – PingTalk Chat App Easter Egg**

Topic: Strings | Difficulty: Easy

PingTalk, a messaging app, has added a playful “Reverse Mode” feature: when a user enables it, the app reverses the ORDER of the words in a message before sending, while keeping each individual word unchanged. Implement this feature.

Input Format:

A single line containing a sentence (words separated by single spaces).

Output Format:

Print the sentence with the order of words reversed.

Constraints:

$1 \leq \text{number of words} \leq 1000$

Sample Input

```
Hello how are you
```

Sample Output

```
you are how Hello
```

Q15**Newsletter Email List Cleanup**

Topic: Arrays / Hashing | Difficulty: Easy

The marketing team at BrightAds imported a bulk subscriber list from several sources, and it now contains duplicate email addresses. Before sending the next newsletter, they need a clean list with duplicates removed, keeping only the FIRST occurrence of each email in its original order.

Input Format:

First line: integer n . Next n lines: one email address each.

Output Format:

Print each unique email on its own line (in original order of first appearance), followed by a final line: "Unique count: X".

Constraints:

$1 \leq n \leq 10^5$

Sample Input

```
6
a@x.com
b@x.com
b@x.com
a@x.com
c@x.com
b@x.com
d@x.com
```

Sample Output

```
a@x.com
b@x.com
c@x.com
d@x.com
Unique count: 4
```

Q16**Tile Sizing – DecorNest Interiors Project**

Topic: Math | Difficulty: Easy

DecorNest, an interior design company, wants to tile a rectangular room using the largest possible square tiles so that no tile ever needs to be cut. Given the room's length and width (in feet), find the side length of the largest such square tile (their GCD), and also find the LCM, which the purchase team uses while planning bulk order batch sizes.

Input Format:

Two space-separated integers L and W (length and width in feet).

Output Format:

Print in the format: "GCD: g, LCM: l".

Constraints:

$1 \leq L, W \leq 10^6$

Sample Input

18 24

Sample Output

GCD: 6, LCM: 72

Q17**Auditorium Seat Preview – EventPro**

Topic: Pattern Printing | Difficulty: Easy

EventPro, an event-management company, is building a quick preview tool for auditorium seating charts. For a given number of rows, the tool must render a right-triangle number pattern representing seat counts, where row i (starting at 1) prints the numbers 1 through i.

Input Format:

A single integer N (number of rows).

Output Format:

Print the pattern described above, one row per line.

Constraints:

$1 \leq N \leq 20$

Sample Input

4

Sample Output

1
1 2
1 2 3
1 2 3 4

Q18**Bank Account Check Digit – Digital Root**

Topic: Math | Difficulty: Easy

SecureBank uses a simple trick to generate a quick check digit for internal test account numbers: repeatedly sum the digits of the number until only a single digit remains (the “digital root”). Write a program that computes the digital root of a given account number.

Input Format:

A single integer N.

Output Format:

Print the digital root of N (a single digit).

Constraints:

$1 \leq N \leq 10^{18}$

Sample Input

9875

Sample Output

2

Q19**Expense Match Finder – SplitEasy App**

Topic: Arrays / Hashing | Difficulty: Medium

SplitEasy, a bill-splitting app, lets a user set a reimbursement target amount and then searches a friend's list of individual expense entries for two expenses that add up EXACTLY to that target, so they can be merged into a single settlement. Given the expense list and the target, print the two expense values (in the order they appear) that sum to it. Assume exactly one valid pair exists.

Input Format:

First line: integer n. Second line: n space-separated integers (expenses). Third line: integer target.

Output Format:

Print the two expense values that sum to the target, space-separated, in the order they appear in the list.

Constraints:

$2 \leq n \leq 10^5$, $1 \leq \text{expense}$, $\text{target} \leq 10^7$

Sample Input

5
40 25 15 60 35
75

Sample Output

15 60

Q20**Smart Parking Slot Allocator – CityMall**

Topic: Simulation | Difficulty: Medium

CityMall has installed a smart parking system with N numbered slots (1 to N), all initially free. Whenever a car arrives, the system must allot the LOWEST-numbered currently free slot ("PARK"); whenever a car leaves, its slot becomes free again ("LEAVE x"). Process a sequence of such commands and, for every PARK command, print the slot number that was allotted.

Input Format:

First line: two integers N and Q (total slots, number of queries). Next Q lines: either "PARK" or "LEAVE x" (x = slot number to free).

Output Format:

For every PARK command, print the allotted slot number on its own line, in order.

Constraints:

$1 \leq N \leq 10^4$, $1 \leq Q \leq 10^4$

Sample Input

```
3 4
PARK
PARK
LEAVE 1
PARK
```

Sample Output

```
1
2
1
```